

Biology Questions Waec Candidate/Private

Question 1

- (a) State one function of each of the following structures: (i) Arteries
- (ii) Capillaries
- (iii) Veins
- (iv) Xylem
- (v) Phloem
- (b) Describe the changes that take place in the composition of human blood as it passes through the: (i) Lungs
- (ii) Liver
- (iii) Kidney
- (iv) Small intestine
- (c) List the cellular components of blood.

Question 2

- (a) What is transpiration?
- (b) (i) Name two types of transpiration.
- (ii) List four factors which affect the rate of transpiration.
- (c) Describe the mechanisms of opening and closing of the stomata.
- (d) State: (i) Two similarities between transpiration and sweating.
- (ii) Two differences between transpiration and sweating.

Question 3

- (a) Explain the following terms: (i) Population
- (ii) Competition
- (iii) Succession
- (b) (i) State three sources of noise pollution.
- (ii) List two ways by which noise pollution can be reduced.
- (c) State three effects each of the following pollutants on humans: (i) Smoke
- (ii) Sewage

Question 4

- (a) Describe the main stages of mitotic division.
- (b) State four ways in which mitosis is important to living organisms.

Question 5

- (a) What are hormones?
- (b) In a tabular form, state four differences between nervous coordination and hormonal coordination in humans.
- (c) State three effects each of: (i) Over-secretion of thyroxin on mammals.
- (ii) Under-secretion of thyroxin on mammals.
- (d) Make a diagram 10-12 cm long of the reflex arc and label fully.

Question 6

- (a) Explain the following modes of nutrition which occur in plants: (i) Autotrophic
- (ii) Chemosynthetic
- (iii) Carnivorous
- (b) Give one example each of a plant which undergoes the mode of nutrition listed in 6(a).
- (c) (i) Describe three ways by which nitrogen is added to the soil for plant use. (ii) State two ways by which nitrogen is lost from the soil.

Question 7

- (a) What is respiration?
- (b) In a tabular form, state four differences between gaseous exchange and aerobic respiration.
- (c) Explain the term residual air.
What is the importance of residual air to mammals?
- (d) State four characteristic features associated with respiratory structures.
- (e) What is oxygen debt?
Outline three activities that can result in oxygen debt.

Question 8

- (a) What is photosynthesis?
List: (i) Four external factors that affect the rate of photosynthesis.
- (ii) Two internal factors that affect the rate of photosynthesis.
- (b) List the major products of the light-dependent stage of photosynthesis.
State the importance of each of the products listed.
- (c) (i) Explain why there are no green plants at the lower depths of some lakes.
- (ii) State why decomposers are important to flowering plants.

Question 9

- (a) What is conservation?
State six factors responsible for the decline of the abundance and variety of wildlife.
- (b) Outline six ways in which the government can improve the situation in 9(a).
- (c) What is Eutrophication?
State two causes of eutrophication.

Question 10

- (a) Describe the carbon cycle.
- (b) State the functions of: (i) Nitrogen-fixing bacteria
- (ii) Nitrifying bacteria
- (iii) Denitrifying bacteria

Question 11

- (a) List three characteristics of living organisms.
- (b) Explain briefly the characteristics listed in 11(a).
- (c) In a tabular form, state four differences between plants and animals.

(d) (i) Name one organism that exhibits both plant and animal features. (ii) State four animal features and two plant features possessed by the organism named in 11(d)(i).

Question 12

- (a) Define the following terms: (i) Elimination
(ii) Excretion
(iii) Secretion
- (b) Describe the digestion of boiled fish in the gut of humans.
- (c) Make a diagram 8 cm – 10 cm long of a longitudinal section of a villus and label fully.

Question 13

- (a) What is osmosis?
- (b) Explain the term turgor pressure.
Describe the effect of turgor pressure in plants.
- (c) In a tabular form, state four differences between osmosis and diffusion.

Question 14

- (a) Define the following terms: (i) Ingestion
(ii) Digestion
(iii) Absorption
(iv) Assimilation
(v) Egestion
- (b) Name the digestive enzymes secreted by the following and state one function of each enzyme: (i) Salivary glands
(ii) Stomach
(iii) Pancreas
(iv) Small intestine

Question 15

- (a) Describe how urine is formed in the kidney.
- (b) List the components of the excretory system.

Question 16

- (a) What is an ecosystem?
- (b) Explain the terms: (i) Autotrophs
(ii) Heterotrophs
(iii) Decomposers
- (c) List the main components of an ecosystem.
State one function of each component listed.

Question 17

- (a) Define the term adaptation.
- (b) Explain the importance of adaptations in organisms.
- (c) List three adaptations each of: (i) Aquatic plants
(ii) Desert plants

Question 18

- (a) What is a habitat?

(b) State four factors that affect the distribution of organisms in a habitat.

Question 19

(a) Define the term mutation.

(b) Explain the difference between a gene mutation and a chromosomal mutation.

(c) List three causes of mutations.

(d) State three effects of mutations on organisms.

Question 20

(a) What is selective breeding?

(b) Explain the difference between inbreeding and outbreeding.

(c) State three advantages and two disadvantages of selective breeding.

Question 21

(a) What is an enzyme?

(b) Explain how enzymes work using the lock-and-key hypothesis.

(c) State four factors that affect enzyme activity.

Question 22

(a) Describe the process of protein synthesis.

(b) Explain the roles of the following in protein synthesis: (i) DNA

(ii) mRNA

(iii) tRNA

Question 23

(a) What is genetics?

(b) Explain the following terms: (i) Genotype

(ii) Phenotype

(iii) Allele

(iv) Homozygous

(v) Heterozygous

(c) Describe the process of meiosis.

State two differences between meiosis and mitosis.

Question 24

(a) What is photosynthesis?

(b) List the main stages of photosynthesis.

(c) Explain the importance of photosynthesis to living organisms.

Question 25

(a) What is biotechnology?

(b) Explain how biotechnology can be used in: (i) Medicine

(ii) Agriculture

(c) List two advantages and two disadvantages of biotechnology.

Question 26

- (a) What is homeostasis?
- (b) Describe the role of the liver in homeostasis.
- (c) Explain the importance of homeostasis in humans.

Question 27

- (a) What is a hormone?
- (b) Name two hormones produced by the pancreas.
State one function of each hormone named.
- (c) Describe the effects of insulin deficiency in humans.

Question 28

- (a) What is immunity?
- (b) Explain the difference between active and passive immunity.
- (c) State four ways in which diseases can be transmitted.

Question 29

- (a) What is reproduction?
- (b) Describe the main stages of the human menstrual cycle.
- (c) List three methods of birth control.
Explain how each method listed works.

Question 30

- (a) Define the following terms: (i) Ecology
(ii) Ecosystem
(iii) Population
(iv) Community
- (b) Describe the flow of energy in an ecosystem.
- (c) Explain the importance of the nitrogen cycle in an ecosystem.

Question 31

- (a) What is ecology?
- (b) Explain the term ecological succession.
- (c) State the difference between primary and secondary succession.

Question 32

- (a) Define the following terms: (i) Pollution
(ii) Pollutant
- (b) List four types of pollution.
State one effect of each type listed.
- (c) Explain the difference between biodegradable and non-biodegradable pollutants.
Give one example of each.

Question 33

- (a) What is a reflex action?
- (b) Describe the pathway of a reflex action.

(c) State the difference between a reflex action and a voluntary action.

Question 34

(a) What is a virus?

(b) Explain the structure of a typical virus.

(c) State three ways in which viruses can be transmitted.

Question 34

1. Which features in the flowering plant correspond to the:

a. Testes?

b. Sperms of a male mammal? (2 marks)

2. With the aid of a labeled diagram, briefly describe the female reproductive system in humans. (12 marks)

3. State:

a. Three differences

b. Three similarities between the male and female reproductive systems in humans. (6 marks)

Question 35

1. Make a diagram 8-10 cm long of the longitudinal section of a mammalian heart and label it fully. (9 marks)

2. Describe briefly pulmonary circulation in mammals. (7 marks)

3. In a tabular form, state four differences between pulmonary circulation and systemic circulation. (4 marks)

Question 36

1. (i) Name the two stages of photosynthesis. (2 marks)

2. (ii) Write chemical equations to represent each stage named in 3(a)(i) above. (6 marks)

3. In a tabular form state:

○ Three vitamins

○ One function

○ One deficiency symptom of each vitamin named. (9 marks)

4. (i) What is roughage? (2 marks)

5. (ii) State one function of roughage. (1 mark)

Question 37

(a) Explain the following terms giving one example in each case:

• (i) Complete dominance;

• (ii) Co-dominance;

• (iii) Mutation. (9 marks)

(b)

• (i) What kind of variation is exhibited by human blood groups? (1 mark)

- (ii) With the aid of genetic diagrams, explain how parents with blood groups 'A' and 'B' respectively can give birth to an offspring with blood group 'O'. (10 marks)

Question 38

- (a) What is excretion? (2 marks)
- (b) Make a diagram 10-12cm long of the mammalian nephron and label fully. (10 marks)
- C) (i) List three diseases of the kidney. (3 marks)
- (ii) List three effects of diseases of the kidney on its function. (3 marks)
 - (iii) Suggest two ways by which the effects listed in 7(c)(ii) above can be remedied. (2 marks)

Question 39

- (a)(i) What is respiration? [3 marks]
- (ii) Explain three ways in which the respiratory surfaces in mammals are adapted to their functions. [6 marks]
- (b) Outline how gases are exchanged in the leaves of flowering plants. [7 marks]
- (c) In a tabular form, state four differences between aerobic and anaerobic respiration. [4 marks]

Question 41

- (a)(i) What is pollution? [2 marks]
- (ii) Name three water pollutants. [3 marks]
- (iii) State five harmful effects of sulphur dioxide in the atmosphere. [5 marks]
- (b)(i) Explain the term population. [3 marks]
- (ii) Explain how a pitfall trap is used to measure the population size of organisms in a habitat. [7 marks]

Question 42

- (a) Explain the application of genetics in marriage counseling giving one example. [5 marks]
- (b) A normal male marries a female carrier of haemophilia. With the aid of a genetic diagram, determine what proportion of their offspring will suffer from the disease. [9 marks]
- (c) State the two laws of inheritance propounded by Gregor Mendel. [6 marks]

Question 43

- (a)(i) What is conservation of natural resources? [3 marks]
- (ii) List five benefits of conservation. [5 marks]
- (b)(i) Name five mineral resources that could be conserved. [5 marks]
- (ii) With appropriate examples, explain the problems associated with conservation of mineral resources. [7 marks]

Question 44

- (a)(i) List three characteristics of living organisms. [3 marks]
- (ii) Explain briefly the characteristics listed in 1(a)(i) above. [6 marks]
- (b) In a tabular form, state four differences between plants and animals. [4 marks]
- (c)(i) Name one organism that exhibits both plant and animal features. [1 mark]
- (ii) State four animal features and two plant features possessed by the organism named in 1(c)(i). [6 marks]

Question 45

- (a) Define the following terms:
- (a) **Elimination:** The removal of undigested food residues (feces) from the body through the anus.
- (b) **Excretion:** The process by which metabolic wastes, mainly nitrogenous compounds like urea, are removed from the body fluids.
- (c) **Secretion:** The release of useful substances (e.g., enzymes, hormones) produced by cells into body fluids or cavities. [6 marks]
- (b) Describe the digestion of boiled fish in the gut of humans. [6 marks]
- (c) Make a diagram 8 cm – 10 cm long of a longitudinal section of a villus and label fully. [8 marks]

Question 46

- 1.(a) Define the following terms:
- (a) elimination
- (b) excretion
- (c) secretion. [6 marks]
- (b) Describe the digestion of boiled fish in the gut of humans. [6 marks]
- (c) Make a diagram 8 cm – 10 cm long of a longitudinal section of a villus and label fully. [8 marks]

Question 47

A survey to determine blood groups was carried out on 250 people living a community. The results are represented in the Table below

Blood group	Percentage (%)
A	8.0
B	14.0
AB	32.8
O	45.2

- (a) (i) Explain the term co-dominance. [2 marks]
- (ii) Calculate the number of individuals with co-dominant blood group. [3 marks]
- (iii) What is the total number of individuals in the Table that are able to donate blood to an accident victim with blood group B? [6 marks]

(b) A man whose blood group is heterozygous A is married to a woman whose blood group is AB. With the aid of a genetic diagram, suggest the possible blood groups of their children. [9 marks]

Question 48

(a) State **five** reasons why animals move from place to place. [5 marks]

(b) State **one** function **each** of the following structures found in plants: [3 marks]

- (i) epidermis:
- (ii) phloem:
- (iii) sclerenchyma:

(c) Give **two** examples **each** of the following types of organisms: [4 marks]

- (i) parasitic plants:
- (ii) saprophytes:

(d) State **two** adaptive features of plants which inhabit salt water swamps. [4 marks]

(e) State **two** factors **each** which result in: [4 marks]

- (i) increase in population density:
- (ii) decrease in population density:

(f) Mention **four** ways in which modern agricultural activities may threaten the survival of species. [4 marks]

(g) State **three** reasons why mitosis is important to living organisms. [3 marks]

(h) State **three** causes of food spoilage. [3 marks]

Question 49

1. What is **classification of living things**? [2 marks]

2. State **four** ways **each** by which the following groups of organisms are of economic importance in their habitats: [8 marks]

- (a) fungi
- (b) plants

3. Complete the table below by stating **five major** phyla of the Kingdom Animalia and give **one** example of **each**. [10 marks]

Phyla of Kingdom Animalia	Example

Question 50

Complete the table below. [10 marks]

Element	Function in plants	Effects of deficiency in plant
Iron		
Molybdenum		
Potassium		
Copper		
Nitrogen		

2. Name three classes of organic foods which are essential constituents in the diet of a mammal. [3 marks]
3. State two roles each of the following structures of the digestive system in humans: [6 marks]
 - Buccal cavity
 - Duodenum
 - Stomach
4. Name the end product of adding dilute hydrochloric acid to sucrose. [1 mark]

Question 51

Relationships **I, II, III, IV, and V** exist among some organisms in an ecosystem. **Use them to answer questions 3(a) to 3(e).**

I: Organism **A** grows on dead organism **B**.

II: Organism **C** feeds on the remnants of food left by organism **D** without affecting organism **D** negatively or positively.

III: Organism **E** provides shelter for organism **F** while organism **F** transports organism **E** towards food.

IV: Organism **G** kills organism **B** for food.

V: Organism **H** feeds on organism **J**, causing **J** a disease.

a. Name the type of relationship in I, II, III, IV, and V. [5 marks]

b. In relationship IV, what is the biological term for each of organisms B and G? [2 marks]

- **B:**
- **G:**

c. Name one example each of organisms B to J. [8 marks]

d. (i) Name the relationship that is most beneficial to the ecosystem. [1 mark]

(ii) Give one reason for the answer in 3(d)(i). [1 mark]

e. (i) Give two examples of organism A. [2 marks]

(ii) Name the group to which organism A belongs. [1 mark]

